

Credit Risk Modeling with WOE-Logistic Regression

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1 Introduction

This project aims to predict whether a borrower will experience serious delinquency within two years using the Kaggle Give Me Some Credit dataset. The model is designed to support credit risk assessment by ranking applicants according to their probability of default.

2 Dataset Overview

The dataset contains approximately 150,000 training samples. The target variable is `SeriousDlqin2yrs`, which indicates whether a borrower experienced serious delinquency within the next two years: The target variable is `SeriousDlqin2yrs`, where 0 indicates no serious delinquency and 1 indicates that serious delinquency occurred within the next two years.

The main variables include `RevolvingUtilizationOfUnsecuredLines` for credit utilization, `age` for borrower age, `NumberOfTime30-59DaysPastDueNotWorse` for the number of 30–59 days past due events, `DebtRatio` for debt burden, `MonthlyIncome` for monthly income, `NumberOfTimes90DaysLate` for the number of 90+ days past due events, and `NumberOfDependents` for the number of dependents.

After removing duplicate records and the ID column, the dataset structure and missing values are summarized using two tables: the Figures 1 presents the output of `df.info()`, while the Figures 2 shows the number of missing values for each variable.

3 Data Cleaning

After removing duplicate records and the ID column, the distributions, outliers, and correlations of numerical variables were inspected to guide data cleaning decisions. The Figures 3 show that many numerical variables are highly skewed and long-tailed. For example, `MonthlyIncome` and `RevolvingUtilizationOfUnsecuredLines` are strongly right-skewed, suggesting that raw continuous variables

```

df.info()
✓ 0.0s

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150000 entries, 0 to 149999
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   SeriousDlqin2yrs                     150000 non-null int64
1   RevolvingUtilizationOfUnsecuredLines 150000 non-null float64
2   age                                   150000 non-null int64
3   NumberOfTime30-59DaysPastDueNotWorse 150000 non-null int64
4   DebtRatio                             150000 non-null float64
5   MonthlyIncome                         120269 non-null float64
6   NumberOfOpenCreditLinesAndLoans      150000 non-null int64
7   NumberOfTimes90DaysLate               150000 non-null int64
8   NumberRealEstateLoansOrLines          150000 non-null int64
9   NumberOfTime60-89DaysPastDueNotWorse 150000 non-null int64
10  NumberOfDependents                    146076 non-null float64
dtypes: float64(4), int64(7)
memory usage: 12.6 MB

```

Figure 1: df.info()

```

df.isnull().sum()
✓ 0.0s

SeriousDlqin2yrs                0
RevolvingUtilizationOfUnsecuredLines 0
age                             0
NumberOfTime30-59DaysPastDueNotWorse 0
DebtRatio                       0
MonthlyIncome                   29731
NumberOfOpenCreditLinesAndLoans 0
NumberOfTimes90DaysLate         0
NumberRealEstateLoansOrLines    0
NumberOfTime60-89DaysPastDueNotWorse 0
NumberOfDependents              3924
dtype: int64

```

Figure 2: df.isnull().sum()

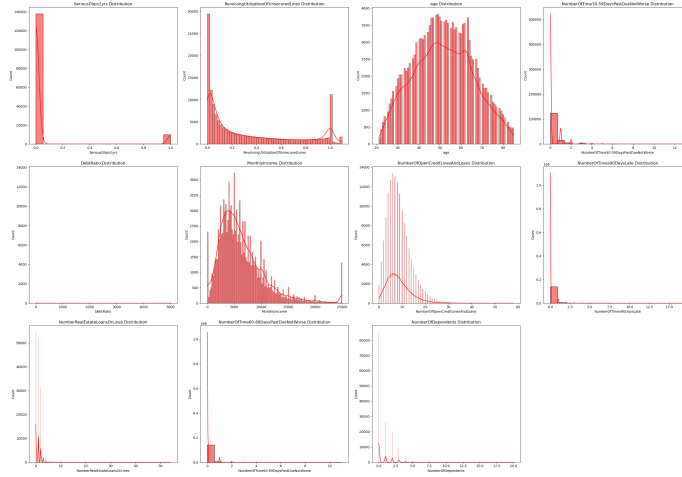


Figure 3: histogram of distribution

may be difficult to interpret directly. Therefore, binning was later applied to transform continuous variables into more interpretable risk groups. The Figures 4 show that credit risk data contains many extreme values, especially in DebtRatio, MonthlyIncome, and RevolvingUtilizationOfUnsecuredLines. These values may represent unusual but possible borrower profiles, but they can distort model estimation if used directly. Therefore, selected continuous variables were capped at the 99th percentile calculated from the training set.

The Figures 5 shows that most predictors have low to moderate pairwise correlations, with no severe multicollinearity observed. Delinquency-related variables and RevolvingUtilizationOfUnsecuredLines show relatively higher positive correlations with the target, which is consistent with credit risk intuition. Based on this analysis, all variables were initially retained for subsequent binning and WOE/IV analysis.

Several cleaning steps were applied based on the exploratory analysis. First, duplicate records and the ID column were removed because they do not provide additional predictive information. Borrowers with ages outside the range of 18 to 85 were also excluded to ensure that the sample reflects a reasonable lending population.

Second, the values 96 and 98 observed in the delinquency count variables were treated as abnormal codes rather than valid delinquency counts. Since replacing these values with zero would incorrectly classify them as borrowers with no delinquency history, these observations were removed from the dataset.

Third, missing values in MonthlyIncome were imputed using the median value calculated from the training set. Median imputation was chosen because income is highly right-skewed and therefore more sensitive to extreme values if the mean is used. The same training-set median was then applied to the validation set to avoid data leakage.

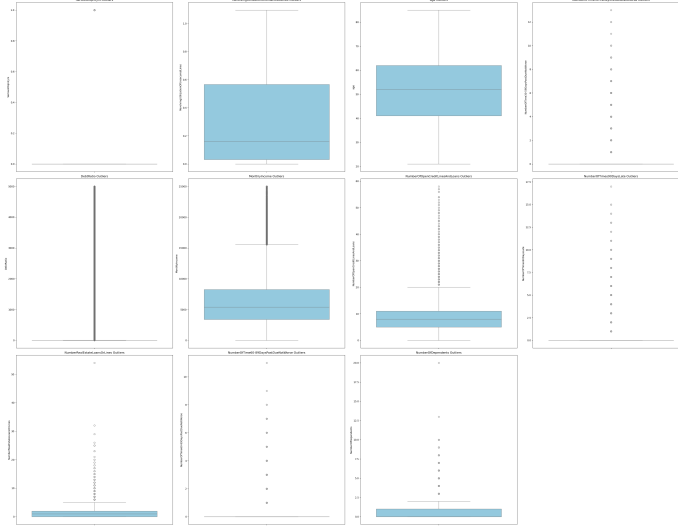


Figure 4: boxplots of outliers

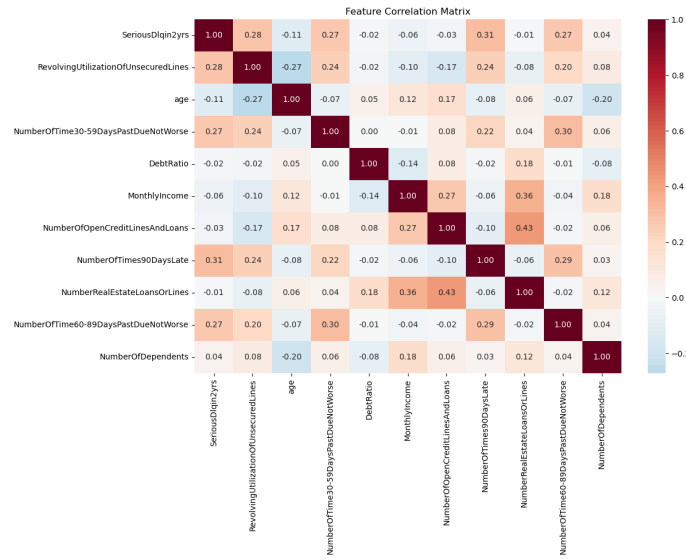


Figure 5: correlation heatmap

Finally, extreme values in `RevolvingUtilizationOfUnsecuredLines`, `DebtRatio`, and `MonthlyIncome` were capped at the 99th percentile calculated from the training set. This treatment reduces the influence of extreme observations while preserving the majority of valid risk information in the data. All preprocessing parameters, including the income median and capping thresholds, were learned only from the training set and then applied to the validation set.

4 Feature Engineering

Feature engineering was conducted to transform raw variables into more stable and interpretable risk features. Since many numerical variables are skewed and contain extreme values, directly using raw continuous variables may reduce model stability and interpretability. Therefore, binning was first applied to convert continuous variables into discrete risk groups. After binning, Weight of Evidence (WOE) transformation and Information Value (IV) analysis were used to encode the variables and evaluate their predictive power.

4.1 Binning

Binning was applied to capture non-linear relationships between predictors and default risk. For continuous variables such as age, `RevolvingUtilizationOfUnsecuredLines`, `DebtRatio`, `MonthlyIncome`, `NumberOfOpenCreditLinesAndLoans`, and `NumberRealEstateLoansOrLines`, a decision-tree-based binning method was used to learn split points from the training set. The tree was constrained by the maximum number of leaf nodes and the minimum sample size in each leaf to avoid overly small or unstable bins.

For delinquency count variables, business-based bins were used because these variables have clear risk meanings. For example, 30–59 days past due was grouped into no delinquency, once, twice, and more than twice. Similar grouping rules were applied to 60–89 days and 90+ days past due variables. All binning rules were learned or defined before being applied to the validation set, so the validation data was processed consistently with the training data.

4.2 WOE and IV

After binning, each bin was transformed using WOE. WOE measures the relative distribution of good and bad borrowers within each bin, allowing categorical risk groups to be converted into numerical values while preserving risk direction. In this project, a lower WOE value indicates a higher-risk group because it contains a relatively larger proportion of bad borrowers.

IV was then calculated to measure the predictive power of each variable. Variables with IV below 0.02 were treated as weak predictors and excluded from the final model. The IV ranking in Table 1 shows that credit utilization, and past delinquency variables are the strongest predictors. This is consistent

with credit risk intuition, as high credit utilization and previous delinquency history are closely related to future default risk.

Table 1: IV Ranking of Binned Variables

Feature	IV	Strength
Rev	1.006	Suspiciously strong
90Days_Bin	0.825	Suspiciously strong
30-59Days_Bin	0.728	Suspiciously strong
60-89Days_Bin	0.556	Suspiciously strong
Age	0.233	Medium
OpenCredit_Bin	0.080	Weak
Income_Bin	0.072	Weak
DebtRatio_Bin	0.067	Weak
RealEstate_Bin	0.056	Weak
Dependents_Bin	0.031	Weak

5 Model Development

After feature engineering, a Logistic Regression model was developed using the WOE-transformed variables. Logistic Regression was selected because it is widely used in credit risk modeling and provides strong interpretability. Compared with more complex black-box models, Logistic Regression allows the direction and magnitude of each variable’s effect on default risk to be more easily understood. This is especially important in credit risk applications, where model decisions often need to be explained to business teams, risk managers, and regulators.

The model estimates the probability that a borrower will experience serious delinquency within the next two years. In this project, the dependent variable is `SeriousDlqin2yrs`, and the independent variables are the selected WOE-transformed features after IV filtering. The model can be expressed as:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1x_1 + \beta_2x_2 + \cdots + \beta_kx_k$$

where p represents the predicted probability of serious delinquency, and $x_1, x_2, \dots, x_k$ represent the WOE-transformed input variables.

The model was trained on the training set and evaluated on the validation set. This setup ensures that model performance is assessed on unseen data rather than on the data used for fitting.

6 Model Evaluation

Since the dataset is highly imbalanced, accuracy alone is not sufficient for evaluating model performance. Therefore, this project uses AUC, KS, and Gini to

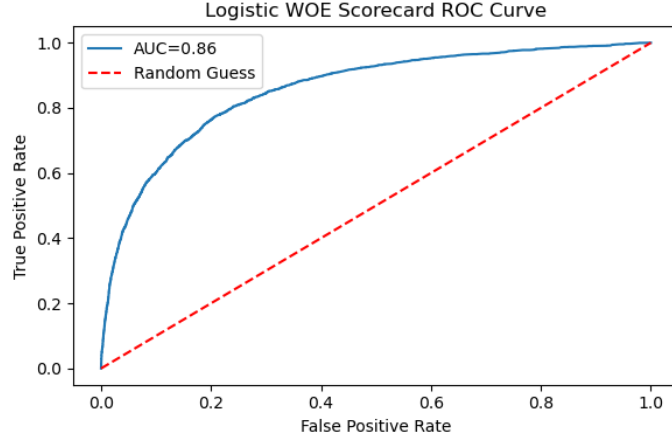


Figure 6: roc curve

evaluate the discriminatory power of the Logistic WOE model.

AUC measures the model’s overall ranking ability, indicating how well the model ranks bad borrowers above good borrowers. KS measures the maximum separation between the cumulative distributions of good and bad borrowers. Gini is derived from AUC and is calculated as $2 \times AUC - 1$. Higher values of these metrics indicate stronger ability to distinguish between high-risk and low-risk borrowers.

Table 2: Model Performance on Validation Set			
Model	AUC	KS	Gini
Logistic WOE Scorecard	0.857	0.567	0.713

As shown in Table 2, the Logistic WOE Scorecard achieves an AUC of 0.857, a KS of 0.567, and a Gini of 0.713 on the validation set. These results indicate that the model has strong discriminatory power and can effectively separate high-risk borrowers from low-risk borrowers.

The ROC curve in Figure 6 further supports this result. The blue curve is clearly above the red random-guessing line, showing that the model performs substantially better than random classification across different thresholds. This suggests that the model can be used to rank borrowers by default risk and support downstream credit approval strategy analysis.

7 Decile and Lift Analysis

To further evaluate the ranking ability of the model, the validation samples were sorted by predicted default probability from high to low and divided into ten

equal-sized groups. Decile 1 represents the highest-risk 10% of borrowers, while Decile 10 represents the lowest-risk 10%.

Table 3: Decile and Lift Analysis on Validation Set

Decile	Total	Bad	Bad Rate	Cum. Bad Capture	Lift
1	2954	1039	35.17%	52.82%	5.28
2	2954	359	12.15%	71.07%	1.82
3	2954	208	7.04%	81.65%	1.06
4	2953	132	4.47%	88.36%	0.67
5	2954	75	2.54%	92.17%	0.38
6	2954	50	1.69%	94.71%	0.25
7	2953	35	1.19%	96.49%	0.18
8	2954	27	0.91%	97.86%	0.14
9	2954	22	0.74%	98.98%	0.11
10	2954	20	0.68%	100.00%	0.10

As shown in Table 3, the highest-risk decile contains 1,039 bad borrowers, with a bad rate of 35.17%. This is much higher than the overall bad rate in the validation set. The lift of the first decile is 5.28, meaning that borrowers in this group are more than five times as risky as the average borrower.

The cumulative bad capture rate also shows strong ranking performance. The top 10% highest-risk borrowers capture 52.82% of all bad borrowers, while the top 20% capture 71.07%. This indicates that the model is effective at ranking borrowers by default risk and concentrating bad borrowers in the highest-risk groups.

The bad rate decreases steadily from Decile 1 to Decile 10, from 35.17% to 0.68%. This monotonic pattern supports the model’s ability to separate high-risk and low-risk borrowers, and provides a useful basis for downstream cutoff strategy analysis.

8 Cutoff Strategy Analysis

Based on the predicted default probabilities, several cutoff strategies were simulated to evaluate how the model could support credit approval decisions. Borrowers were sorted by predicted risk from high to low, and the highest-risk 10%, 20%, and 30% of borrowers were rejected respectively.

Table 4: Cutoff Strategy Analysis on Validation Set

Reject Rate	Approval Rate	Bad Capture Rate	Approved Bad Rate	Rejected Bad Rate
10%	90%	52.82%	3.49%	35.18%
20%	80%	71.07%	2.41%	23.67%
30%	70%	81.65%	1.75%	18.12%

As shown in Table 4, rejecting the top 10% highest-risk borrowers captures

52.82% of all bad borrowers while maintaining a 90% approval rate. The bad rate among approved borrowers is reduced to 3.49%, while the rejected group has a bad rate of 35.18%.

When the reject rate increases to 20%, the model captures 71.07% of bad borrowers and reduces the approved bad rate to 2.41%. A more conservative strategy that rejects the top 30% highest-risk borrowers captures 81.65% of bad borrowers and further lowers the approved bad rate to 1.75%, but the approval rate decreases to 70%.

This analysis shows the trade-off between risk control and business volume. A higher reject rate can capture more bad borrowers and lower the risk of the approved population, but it also reduces the approval rate. Therefore, the final cutoff should depend on the institution’s risk appetite and business objectives.

9 Conclusion and Limitations

This project developed an interpretable credit risk model using decision-tree-based binning, WOE transformation, IV feature selection, and Logistic Regression. The model achieved strong performance on the validation set, with an AUC of 0.857, a KS of 0.567, and a Gini of 0.713. The Decile and Lift analysis further showed that the model can effectively rank borrowers by default risk, with the highest-risk 10% of borrowers capturing 52.82% of all bad borrowers.

The cutoff strategy analysis demonstrates how the model can support credit approval decisions. For example, rejecting the top 20% highest-risk borrowers captures 71.07% of bad borrowers and reduces the approved bad rate to 2.41%. This shows that the model can provide practical value by helping balance risk control and approval volume.

However, this project also has several limitations. First, the official Kaggle test set does not provide observable target labels, so model validation was conducted using a stratified hold-out validation set from the training data. Second, this project focuses on offline model development and does not include production-level monitoring such as PSI, probability calibration, reject inference, or post-launch vintage analysis. In a real credit risk environment, these additional analyses would be required before deployment.

Overall, the results show that the WOE-Logistic model provides a strong and interpretable framework for credit risk assessment. It can identify high-risk borrowers, support cutoff strategy design, and provide insights that are consistent with credit risk intuition.